The 280 EUR Homelab Starter Kit

From bare parts to your first self-hosted service
in 4 weekends. Zero recurring SaaS bills.

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ZEROCLAW

self-hosted AI stack - @zeroclaw.build

Why this exists

You are paying 15-100 EUR per month for SaaS tools that could run on hardware you own. A tiny 4-core mini-PC with 16 GB RAM does the job of a mid-tier cloud instance - and pays for itself in 3-6 months.

This is not a comprehensive homelab bible. It is the exact parts list, the exact setup sequence, and the exact first services you should run - the plan I wish I had 12 months ago.

Who this is for:

- You have paid 20+ EUR/mo for at least 2 SaaS tools in the last year
- You are comfortable with a terminal (SSH, apt, systemctl)
- You do not need 24/7 five-nines uptime
- You want to own the stack, not rent it

What you get in this kit:

- Exact 280 EUR parts list (with links + why each)
- 4-weekend setup plan, one weekend at a time
- 5 self-hosted services worth running first
- Monthly cost breakdown (spoiler: 8-13 EUR)
- The pitfalls I learned the hard way

The 280 EUR Shopping List

Prices are Amazon.de / eBay average as of July 2026. Buy used where flagged - the savings are real and the risk is minimal on these parts.

Part	Price	Why this specific one
Mini-PC: Intel N100 or N305 (e.g. Beelink EQ12, GMKtec)	170 EUR	4-core Alder Lake-N, 6W TDP, always-on friendly. N305 has 8
16 GB DDR4 SO-DIMM (Crucial CT16G4SFRA32A)	35 EUR	16 GB is the sweet spot. 32 GB is overkill for weekend 1. Upgr
500 GB NVMe SSD (Crucial P3 or WD Blue SN580)	45 EUR	Boot + LXC storage. NVMe over SATA - the price gap has clos
USB-C PD charger 65 W (if not included)	15 EUR	Most mini-PCs ship with barrel-jack + brick. USB-C PD is porta
USB-Stick 32 GB (for Proxmox install)	8 EUR	Any decent stick. You will use it for backups later too.
Ethernet cable Cat 6, 2 m	7 EUR	WiFi works but Ethernet doubles your throughput and drops yo
TOTAL	280 EUR	One-time. No recurring cost.

Buy-used opportunities: The mini-PC drops to 100-130 EUR on eBay if you accept 1-2 years of use. RAM and NVMe are typically fine used. Never buy a used PSU or SSD without a warranty.

Skip these purchases (for now): Rack, PDU, UPS, KVM, second machine. You can add them in month 6 if you actually need them. Most people do not.

4-Weekend Roadmap

Each weekend is 3-5 hours. You can compress this to 2 weekends if you already know Linux, or stretch to 6 weekends if you have a family and 2-hour windows.

Weekend	Focus	Outcome
1	Hardware + Proxmox install	Proxmox web UI reachable at 192.168.1.x:8006
2	First LXC + Docker inside	docker ps returns something. Portainer accessible.
3	First real service (pick 1 of 5)	One SaaS-replacement running. You use it daily.
4	Monitoring + backup basics	Uptime alerts to Telegram. Daily vzdump to USB.

Weekend 1 - Proxmox on bare metal

Step 1: Prepare the Proxmox installer

Download the Proxmox VE 9 ISO from proxmox.com/downloads (~1.3 GB).

On your Mac / Linux: write it to the USB stick with Balena Etcher or:

```
sudo dd if=proxmox-ve_9.iso of=/dev/sdX bs=4M status=progress
```

(Replace /dev/sdX with your actual USB device. Wrong disk = data gone.)

Step 2: Boot the mini-PC from USB

Plug USB + Ethernet + display. Hit BIOS key (usually F2/F7/Del) at boot.

BIOS settings to check:

- Secure Boot: OFF
- Boot mode: UEFI (not Legacy)
- Virtualization: ON (VT-x + VT-d if present)
- Boot order: USB first

Step 3: Proxmox install choices

Target disk: your NVMe. Filesystem: ext4 (simpler than ZFS for beginners).

Set a strong root password. Note the IP shown at the end.

Timezone: your actual timezone. Hostname: something short like 'pve'.

Step 4: First login

From your Mac / Linux browser:

```
https://192.168.1.x:8006 (accept self-signed cert)
```

Login: root / your-password. Realm: Linux PAM.

Skip the subscription nag: Proxmox shows a subscription warning. You can ignore it, or run the free-repo fix once. Do NOT install Enterprise repos without a subscription - your updates will silently break.

Weekend 2 - First LXC + Docker

Why LXC not VM

LXC containers share the host kernel. 3x less RAM, 5x faster boot, and you can nest Docker inside them. VMs are for Windows or when you need a completely isolated kernel (rare on a homelab).

Step 1: Download a Debian template

In Proxmox web UI: local (pve) - CT Templates - Templates - debian-12-standard.

Wait 60 seconds. It downloads ~300 MB.

Step 2: Create your first LXC

Right-click your node - Create CT. Recommended settings:

- CT ID: 100 (100-199 are yours to use freely)
- Hostname: docker-host
- Template: debian-12-standard
- Root disk: 20 GB (thin-provisioned)
- CPU: 2 cores
- Memory: 2048 MB / Swap: 512 MB
- Network: DHCP, unprivileged, nesting=1

Step 3: Install Docker inside the LXC

SSH into the CT (or use Proxmox console):

```
apt update && apt install -y ca-certificates curl

curl -fsSL https://get.docker.com | sh

docker run hello-world
```

If hello-world runs, Docker is up.

Step 4: Portainer for a GUI

```
docker run -d -p 9000:9000 -p 9443:9443 --name portainer \
--restart=always -v /var/run/docker.sock:/var/run/docker.sock \
-v portainer_data:/data portainer/portainer-ce
```

Browse to https://192.168.1.x:9443. Set admin password. Done.

Weekend 3 - First service worth running

Pick ONE. Do not try all five in weekend 3. The point is to prove that you can go from 'zero' to 'daily user of my own service' in a single weekend.

Option A: Vaultwarden (Bitwarden clone)

Replaces: Bitwarden Premium (10 USD/mo), 1Password (36 EUR/yr)

Setup: 15 min. Docker compose file, add reverse proxy, done.

Why start here: password managers are daily-used, low-risk, and Vaultwarden is rock-solid.

Option B: Immich (self-hosted Google Photos)

Replaces: Google One (2 EUR/mo for 100 GB), iCloud+ (3 EUR/mo)

Setup: 30 min. Needs more storage than 500 GB if you have >5 years of photos.

Why: face recognition, timeline UI, and mobile app that just works.

Option C: Paperless-ngx (document archive + OCR)

Replaces: Evernote Personal (15 EUR/mo), Notion for docs (10 USD/mo)

Setup: 30 min. Add a scanner or scan-to-folder.

Why: keeps every invoice, contract, warranty searchable forever.

Option D: n8n (Zapier alternative)

Replaces: Zapier Starter (20 USD/mo), Make Core (9 EUR/mo)

Setup: 20 min. Free-tier LLM router via Ollama = free Zapier-with-AI.

Why: automations that hit APIs, webhooks, and LLMs without seat licenses.

Option E: Home Assistant (smart-home hub)

Replaces: SmartThings, Alexa Routines, Google Home paid features

Setup: 45 min (VM, not LXC - needs full kernel access for USB dongles).

Why: local-first, no cloud dependency, works when internet is down.

Weekend 4 - Monitoring + backup

Skip this weekend and you will lose data within 3 months. This is not the fun weekend - it is the weekend that makes weekends 1-3 not have been wasted.

Step 1: Uptime alerts to Telegram (30 min)

Create a Telegram bot via @BotFather. Get your chat_id via @userinfobot. Deploy Uptime Kuma in a Docker container:

```
docker run -d --name uptime-kuma -p 3001:3001 \
-v uptime-kuma:/app/data --restart=always \
louislam/uptime-kuma:1
```

Add monitors for: Proxmox host, each LXC, your services. Configure Telegram notifier. You now get pings the second something goes down.

Step 2: Automated backups (45 min)

In Proxmox: Datacenter - Backup - Add. Storage: local. Schedule: daily 03:00.

Select all CTs and VMs. Mode: snapshot. Compression: zstd.

Retention: 7 daily + 4 weekly + 3 monthly.

USB backup pitfall: Consumer USB sticks fail after 3-8 weeks of continuous writes. Rotate 2 sticks and swap monthly, or invest in a proper NAS by month 6.

Step 3: Off-site copy (15 min)

One backup = no backup. Rclone to Backblaze B2 (0.005 USD/GB/mo) for critical data. 50 GB of encrypted config + password-DB = 25 cents per month.

Monthly cost breakdown (real numbers)

Line item	Amount	Notes
Electricity (24/7)	3-5 EUR	10 W avg x 24 h x 30 d x 0.35 EUR/kWh
Internet (marginal)	0 EUR	Already paying it. Traffic is negligible.
Off-site backup	0.25 EUR	50 GB in Backblaze B2
Domain (optional)	0.50 EUR	6 EUR/yr for a .dev or .build TLD
Dynamic DNS	0 EUR	DuckDNS or Cloudflare API - both free
Cloudflare Tunnel	0 EUR	Free tier - no public IP needed
TOTAL	4-6 EUR	vs 47+ EUR/mo you were paying

Break-even math: 280 EUR hardware / 42 EUR saved per month = 6.7 months. Everything after month 7 is pure savings, and the hardware is still going strong at year 5.

Pitfalls I learned the hard way

These are the mistakes that cost me weekends. Do not repeat them.

1. Never trust a single USB backup

Consumer USB sticks silently go read-only after 3-8 weeks of daily writes. My first backup stick died 21 days in, and I only noticed a month later when I needed to restore. Use 2 sticks, rotate, and monitor backup-success actively.

2. Do not run bleeding-edge kernels

Proxmox often ships opt-in preview kernels. Do not switch until they hit LTS. I lost a full day to a memory-management bug in a preview kernel that froze the host every ~40 hours.

3. Do not expose services directly

Never open ports 80/443 straight to your LXC. Use Cloudflare Tunnel (free) or Tailscale Funnel. Direct port-forwards get you scanned within 2 hours and brute-forced within 24.

4. Do not skip the first-week hardening

SSH: keys only, no passwords, non-standard port. Fail2ban for anything internet-facing. Automatic security updates on Proxmox host AND every LXC. 15 minutes of setup, prevents 80% of incidents.

5. Do not treat homelab as production

If your family relies on your Immich for photo access, they WILL yell when it is down. Either accept the SLA (yours is 'best-effort') or keep Google Photos as backup for anything family-critical for the first 6 months.

What is next

Once weekend 4 is done and you have used your service for 4 weeks, you are ready for the next tier. Here is what I would tackle in month 2 and beyond.

Month 2: Add a second service

Pick a second option from weekend 3 that you skipped. The learning curve is 10x flatter the second time - most steps are muscle memory.

Month 3: Automate the boring stuff

n8n + Ollama = free Zapier with local LLMs. Automate: bill categorization, receipt extraction, news filtering, social media cross-posting. Anywhere you copy-paste twice a week is automatable.

Month 4-6: Multi-node or NAS

By month 6 you will hit storage limits or want redundancy. Add a second mini-PC (Proxmox cluster) or a Synology / Terramaster NAS. Both cost about 300-500 EUR.

Month 6+: The interesting stuff

Self-hosted AI (Ollama + qwen2.5:14b runs comfortably on a used 3060 12 GB), Home Assistant with Matter, self-hosted analytics (Plausible), the works. This is the tier where homelab stops being a project and starts being infrastructure you rely on.

Want more? Follow the build.

I document my self-hosted AI stack (ZeroClaw) in the open - the wins, the freezes, the cost breakdowns. Same rules as this PDF: real numbers, no marketing.

- Instagram: @zeroclaw.build
- X: @zeroclaw_build
- Bluesky: @zeroclawbuild.bsky.social

This PDF is v1.0 - July 2026. If you spot mistakes or want more depth on any section, DM me on any of the accounts above.